

Generative Adversarial Networks

Learn to generate by playing a game against a critic

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ES 667 · Deep Learning · IIT Gandhinagar

From a latent noise vector to a fake image

Recall the generative task

We want to turn simple noise into a realistic sample:

$$z \sim p(z) \xrightarrow{G_\theta} \tilde{x}$$

We need a **training signal** that says whether the sample $\tilde{x}$ looks like real data — but for a freshly drawn z there is **no target image** to compare against.

so where does the loss come from?

Why not use only pixel reconstruction?

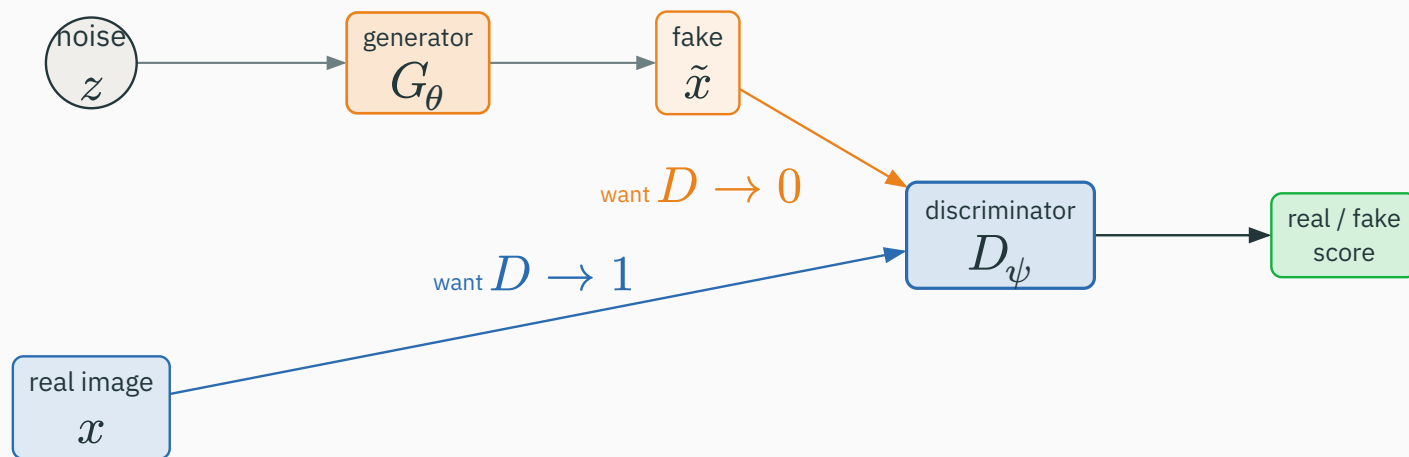
The obvious idea — minimize pixel distance to some target — fails twice:

- a newly sampled z has **no** paired target image to reconstruct;
- even with a target, pixel distance rewards the **average** of plausible images, giving **blurry** results (exactly the VAE symptom from L19).

we need a signal for **realism**, not for pixel-wise proximity

The two-player setup

Turn generation into a game between two networks:



the generator makes fakes; the discriminator scores real vs fake — neither loss makes sense alone

Two networks with opposing goals:

G_θ — **the counterfeiter**. Maps simple noise z to samples $\tilde{x}$; wants the detective to call them real.

D_ψ — **the detective**. Separates real data from generated samples; wants to catch every fake.

their **competition** supplies the learning signal — a learned notion of realism

A discriminator is a classifier

The discriminator is an ordinary binary classifier with a probability output:

$$D_{\psi(x)} \in (0, 1) \approx P(\text{real} \mid x)$$

label real examples **1**, generated examples **0** — the one twist is that the negative class keeps changing

Push D up on real data and down on fakes:

$$\max_{\psi} \mathbb{E}_{x \sim p_{\text{data}}} \log D_{\psi}(x) + \mathbb{E}_{z \sim p(z)} \log \left(1 - D_{\psi}(G_{\theta}(z)) \right)$$

the first term rewards calling real data real; the second rewards calling fakes fake

The generator's original objective

Freeze D ; change the fakes so the detective calls them real:

$$\min_{\theta} \mathbb{E}_{z \sim p(z)} \log \left(1 - D_{\psi}(G_{\theta}(z)) \right)$$

Only the **second** term of the discriminator objective depends on θ . The generator has no access to real data — it only sees the discriminator's verdict on its own samples.

Stack the two objectives into one value function:

$$\min_G \max_D V(D, G)$$

$$V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

the detective **maximizes**, the counterfeiter **minimizes** — the same scalar

The game plays out as an **alternation**, never a single descent:

1. fix D , nudge the **fakes** toward regions D currently calls real;
2. fix G , move the **boundary** to reject those fakes again.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/gan-minimax-dance

Step the minimax dance: watch generated samples drift toward the data while the discriminator boundary chases them — both players keep moving.

How GAN training actually works

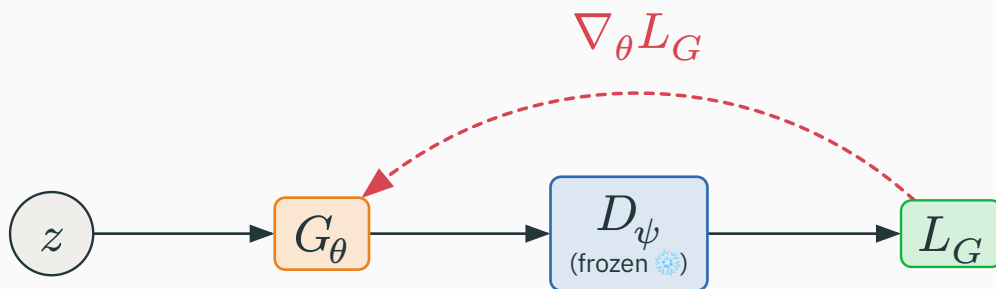
Each step updates the two networks in turn:



update D on a fresh minibatch, then update G on fresh noise — then repeat

The gradient path to the generator

The generator never touches real data; its signal arrives **through** the discriminator:



$$z \rightarrow G_{\theta(z)} \rightarrow D_{\psi}(G_{\theta(z)}) \rightarrow L_G$$

gradients pass through the **frozen** D into G ; D 's weights are held fixed, but its computation stays in the graph

Early on, the detective easily spots fakes, so $D(G(z)) \approx 0$:

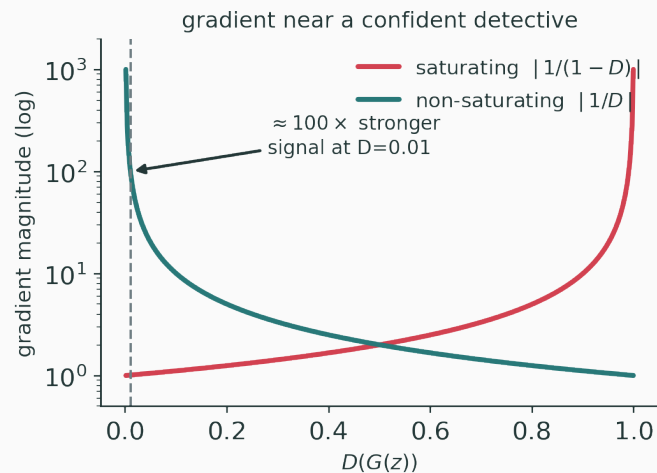
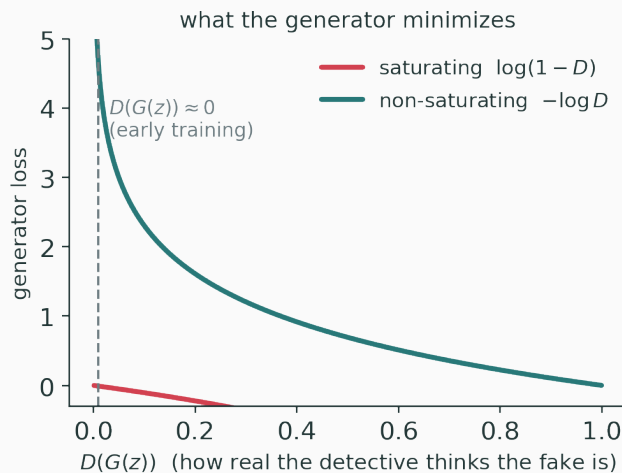
$\log(1 - D(G(z)))$ is very **flat** near $D = 0$

The original $\log(1 - D)$ generator objective gives a **tiny gradient** exactly when the generator is worst and needs the **strongest** push. The signal vanishes early.

Non-saturating generator loss

Keep the same goal — make fakes look real — but flip to a loss with a healthy gradient:

$$L_G = -\mathbb{E}_{z \sim p(z)} \log D_{\psi}(G_{\theta}(z))$$



Take a confident detective: $D(G(z)) = 0.01$.

Saturating $\log(1 - 0.01) = \log 0.99 \approx -0.01$.

Gradient magnitude $1/(1 - D) \approx 1.01$ — nearly flat.

Non-saturating $-\log 0.01 \approx 4.61$.

Gradient magnitude $1/D = 100$ — a strong push.

$\approx 100 \times$ **stronger gradient at $D = 0.01$ — same direction, far more useful signal**

If the generated and real distributions matched **perfectly**, no detective could beat a coin flip:

$$p_G = p_{\text{data}} \implies D(x) = \frac{1}{2} \text{ everywhere}$$

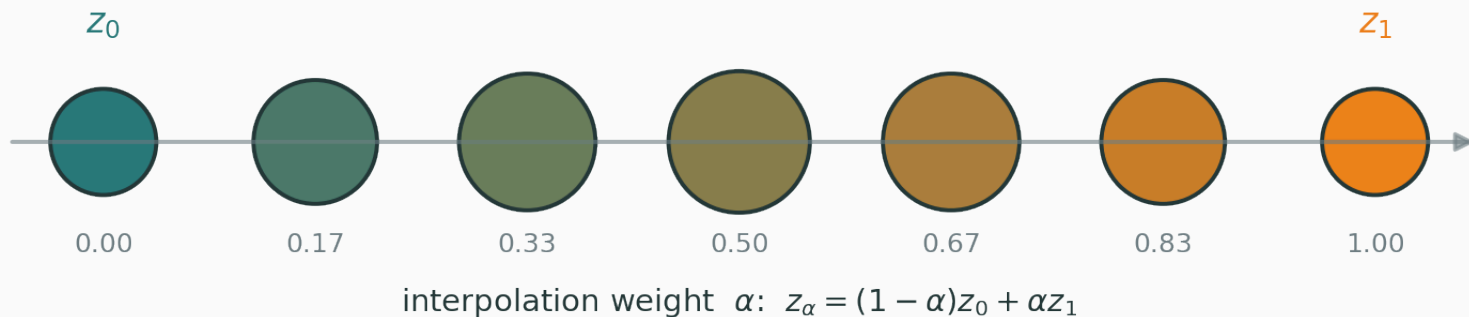
the detective is reduced to guessing — the strongest possible evidence that the fakes match the data

A common misreading of $D = \frac{1}{2}$:

$D(x) = \frac{1}{2}$ at equilibrium does **not** mean every generated image is average or blurry. It means the **two distributions are indistinguishable** — the detective cannot separate them, not that samples are bland.

a statement about **distributions**, not about any single image

Sample two codes, interpolate, and generate along the path:



smooth semantic change is evidence that G has **organized** the noise space into a usable map

Why GANs can be hard to train

Unlike ordinary supervised learning, the loss surface itself **keeps moving**:

Supervised — the target is **fixed**; you descend a stationary loss.

Adversarial — the opponent **learns too**; the objective shifts under you every step.

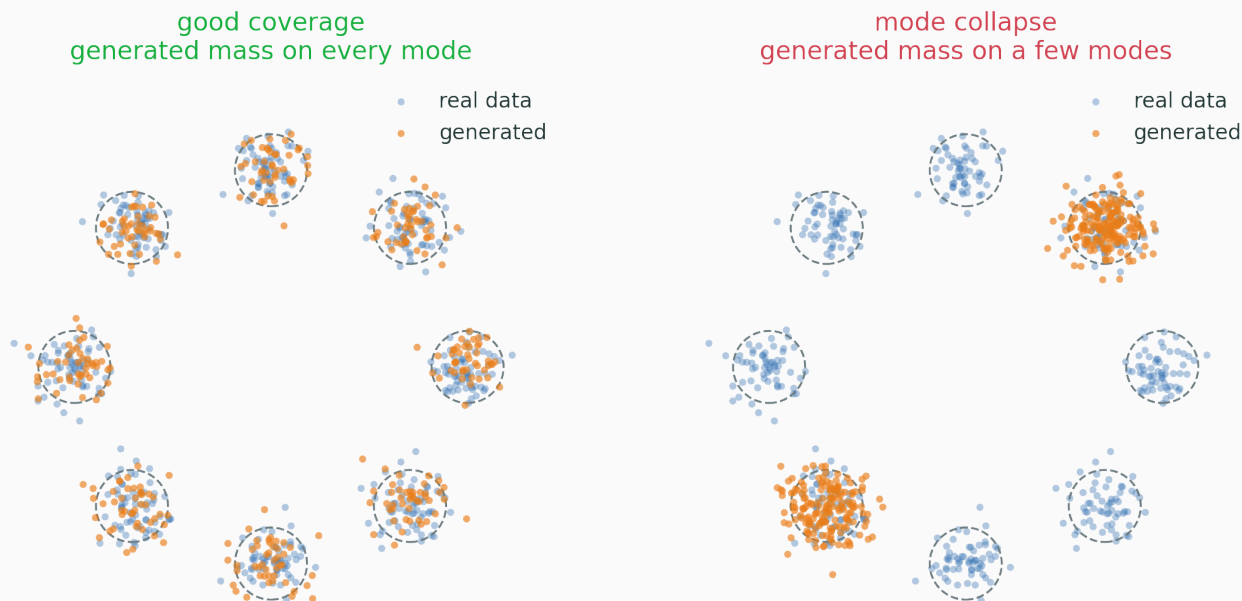
game optimization $\neq$ minimizing one fixed function

If the detective becomes **perfect**, it separates real from fake with total confidence:

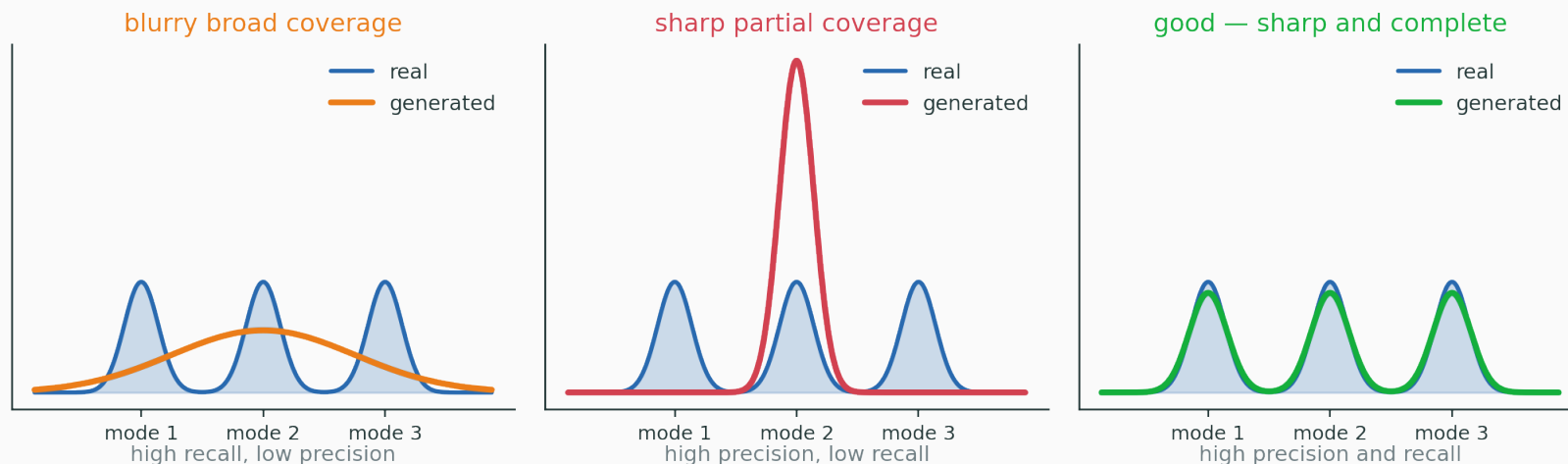
$$D(x) \approx 1 \text{ on real, } D(G(z)) \approx 0 \text{ on fake}$$

A perfect discriminator leaves the generator with **flat, uninformative** gradients — the game stalls because the critic gives no direction.

The generator finds a few outputs that fool the **current** detective, and repeats them:



Sharpness and coverage are **different** axes of quality:



broad-but-blurry, sharp-but-partial, and sharp-and-complete are three genuinely distinct behaviours

The two players can **chase** each other around rather than converge:



training curves alone can mislead — a lower D loss is not a better generator

A grab-bag of tricks that make the game better-behaved:

- **architecture** — convolutional up/down-sampling (DCGAN);
- **balance** — tuned update schedules and regularization;
- **better critics** — distance-like objectives (Wasserstein intuition);
- **monitor samples** — inspect generated grids, not only losses.

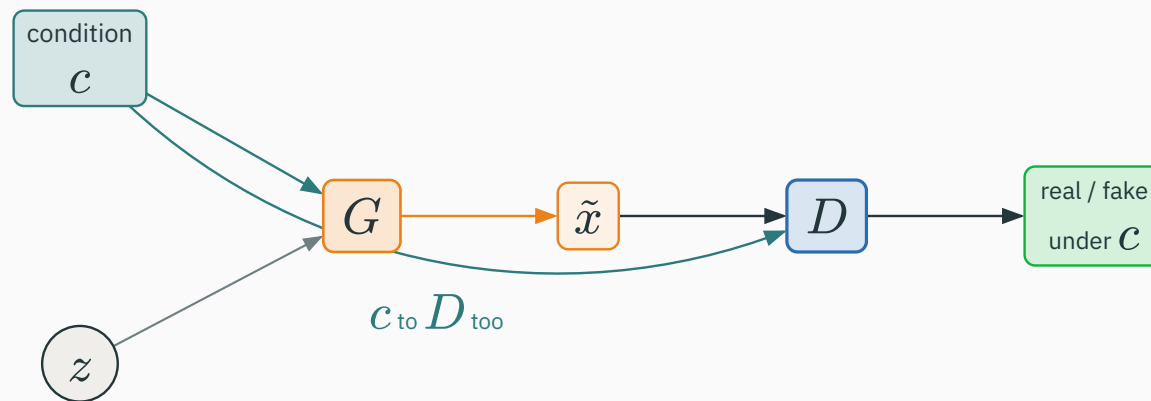
none is a silver bullet — GAN training is a craft as much as an algorithm

A binary detective saturates; a **critic** that scores “how real” gives a smoother signal:

WGAN replaces the saturating classifier with a critic and measures an **earth-mover** distance between distributions. Read it as a response to **unstable geometry**, not a fourth objective to memorize.

we return to the earth-mover picture in Part V

Feed a **condition** c so generation becomes controllable:



$$z, c \xrightarrow{G} \tilde{x}, \quad D(x, c)$$

c can be a class, a caption, a pose, or a segmentation map

GANs in the generative-model landscape

	VAE	GAN
training signal	likelihood bound (ELBO)	learned adversarial critic
inference encoder	yes	not required
typical trade-off	coverage, smoother samples	sharpness, instability / collapse
density estimate	explicit approximate model	implicit sampler

a probabilistic bound versus a learned game — two very different sources of gradient

GANs shine where structure is **paired** or **conditional** and a fast single pass matters:

- **image-to-image translation** (day $\rightarrow$ night, sketch $\rightarrow$ photo);
- **super-resolution** and restoration;
- **domain adaptation** with an adversarial alignment loss.

strong conditional signal + one-shot sampling = a natural GAN fit

Diffusion (next lecture) trades one adversarial pass for **many** stable denoising steps:

- **coverage** — a stationary regression target avoids mode collapse;
- **stability** — no moving opponent, so training is far more reliable;
- **cost** — the price is **many** sampling steps instead of one.

better coverage and stability, at a sampling-time cost

“If samples look **sharp** but all resemble the **same** face, which failure mode is this?”

and — would a discriminator loss **alone** have diagnosed it?

mode collapse — and no, sharp fakes can keep D loss looking healthy

One sentence to carry forward:

GANs learn by fooling a critic; diffusion learns by reversing noise.

Part V now derives the game's optimum, the losses per minibatch, and the failure modes in detail

Detailed adversarial training

For real data label $y = 1$ and generated data label $y = 0$:

$$L_D = -[y \log D(x) + (1 - y) \log(1 - D(x))]$$

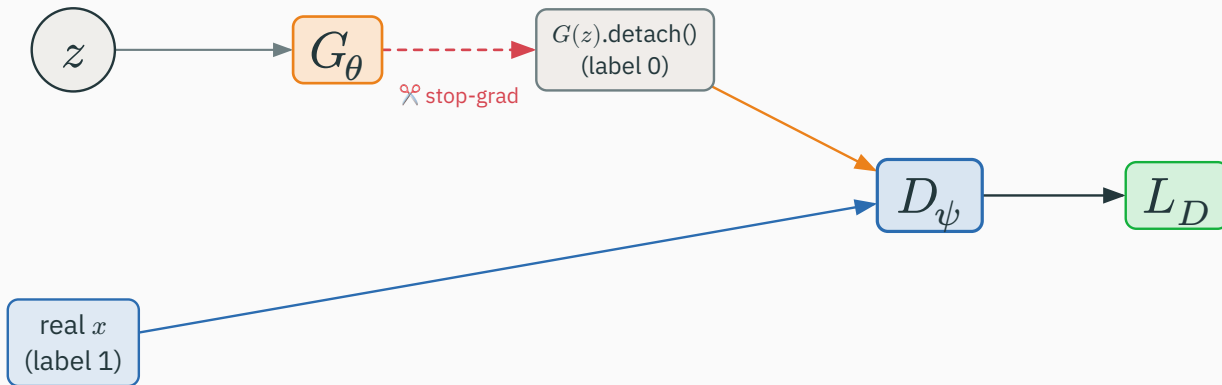
Concretely, if $D(x_{\text{real}}) = 0.9$ and $D(G(z)) = 0.2$:

$$L_D = -(\log 0.9 + \log 0.8) \approx -(-0.105 - 0.223) = \mathbf{0.329}$$

ordinary classification — the twist is that the **negative class** moves as G changes

Average the BCE over a batch of B real images and B fakes:

$$L_D = -\frac{1}{B} \sum_{i=1}^B \log D(x_i) - \frac{1}{B} \sum_{i=1}^B \log(1 - D(G(z_i)))$$



when updating D , the fakes are **detached** — no gradient leaks back into G

Now update G with the non-saturating loss, discriminator frozen:

$$L_G = -\frac{1}{B} \sum_{i=1}^B \log D(G(z_i))$$

The discriminator's weights are **frozen**, but its computation **remains in the graph** — that is how G receives a direction for changing its pixels.

The generator is not trained on “fake” labels

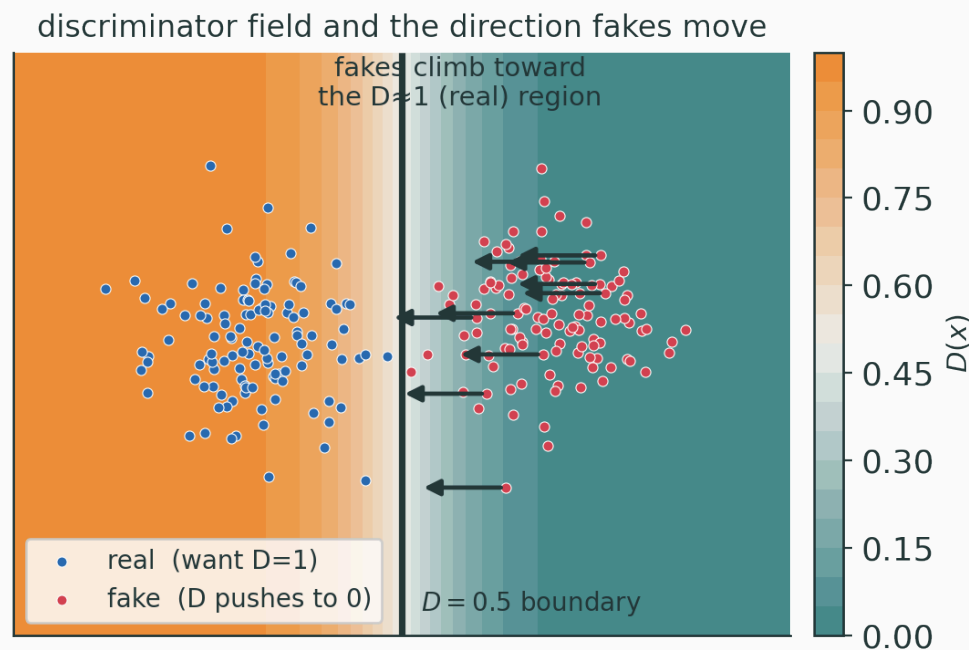
A subtle but crucial point about what G optimizes:

```
update D:  real      -> 1 ,    generated -> 0  
update G:  generated -> make D output 1
```

G does **not** become a classifier — it changes its images so D 's learned features regard them as real

The discriminator's boundary moves

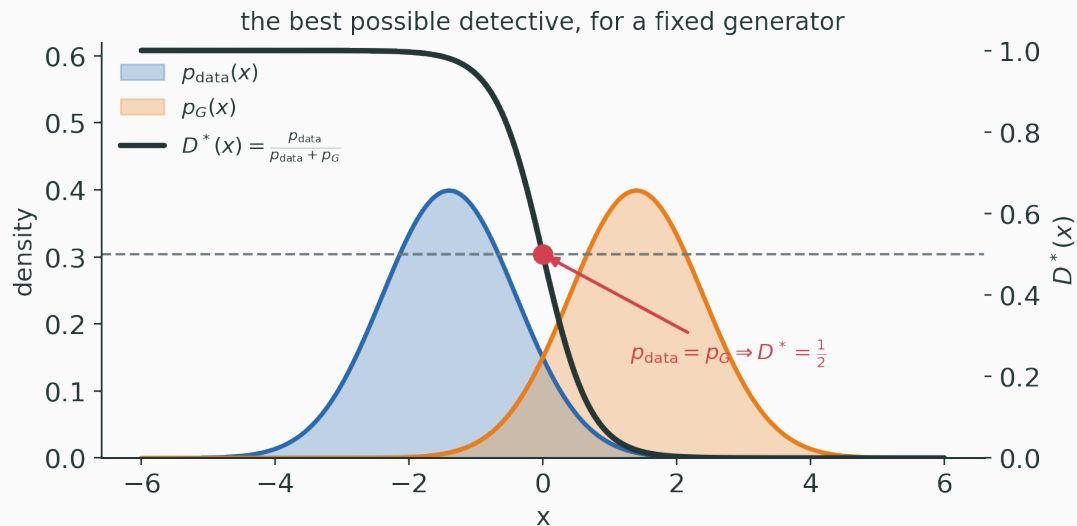
Two real-ish clusters and a fake cluster: after each update the picture shifts.



this moving-boundary picture is worth more than a formal convergence proof on a first pass

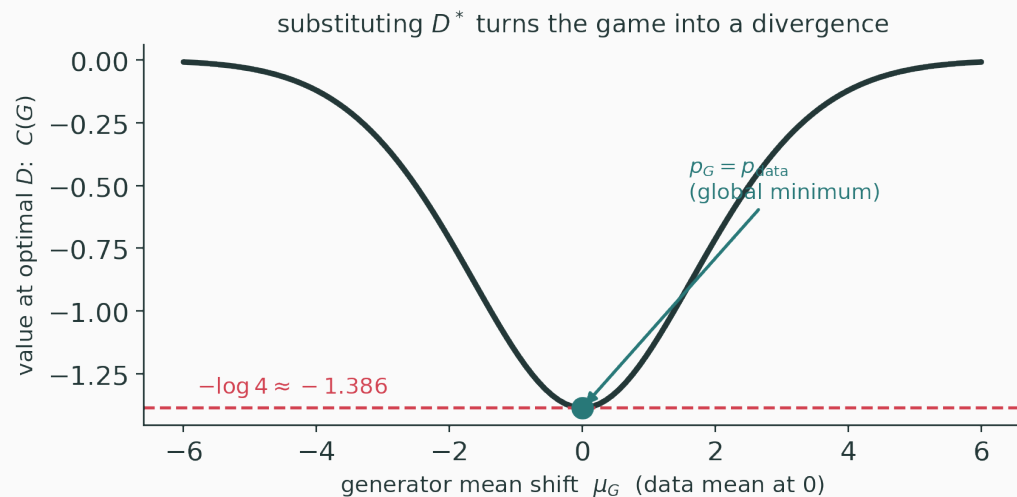
For a **fixed** generator, the optimal discriminator has a closed form:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)}$$



Substituting D^* back turns the game value into a **divergence** between distributions:

$$C(G) = -\log 4 + 2 \cdot \text{JSD}(p_{\text{data}} \| p_G)$$



Strip away the algebra and one idea remains:

GAN training tries to match distributions — not to pair each fake with one real target.

the minimum $-\log 4 \approx -1.386$ is reached exactly when $p_G = p_{\text{data}}$, where $\text{JSD} = 0$

When the detective assigns $D(G(z)) \approx 0$ everywhere:

- the original minimax generator loss **flattens** — a vanishing gradient;
- the non-saturating loss changes the **gradient behaviour** while keeping the **same desired direction**.

same target, different curvature — the fix is about gradient magnitude, not about goals

The whole training loop in five lines:

```
repeat:  
  x <- real minibatch ; z <- random noise  
  update discriminator using x and G(z).detach()  
  z <- fresh random noise  
  update generator using -log D(G(z))
```

the two updates use **different** colours of parameter — D frozen for one, G frozen for the other

One D step per G step is **common**, not a law:

- some critic objectives take **several** discriminator updates per generator update, to keep the signal informative;
- others balance learning rates or add regularization instead.

a first example of why **game** optimization is more delicate than minimizing one fixed loss

Supply the condition c to the generator **and** the discriminator:

$$G(z, c) \rightarrow \tilde{x}, \quad D(x, c) \rightarrow \text{real} / \text{fake}$$

- for **digit generation**, c is a class label;
- for **image translation**, c is an input image, pose, or segmentation map.

D must now judge realism **and** agreement with c — a fake-but-mismatched sample is still rejected

Mechanism	What it does
latent z	chooses one diverse sample
condition c	specifies the desired content
discriminator / critic	judges realism under c

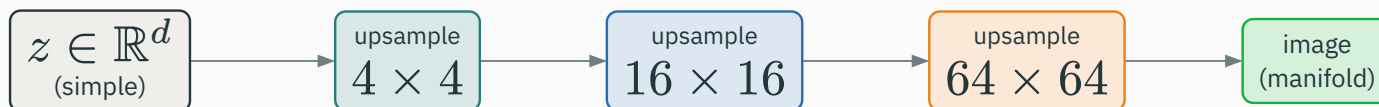
many different z values can produce different **valid** samples for the **same** c

The practical pattern that first made GANs train reliably on images:

- **convolutional** up / down-sampling (no large dense hidden layers);
- careful **normalization** choices;
- **progressively** build spatial resolution.

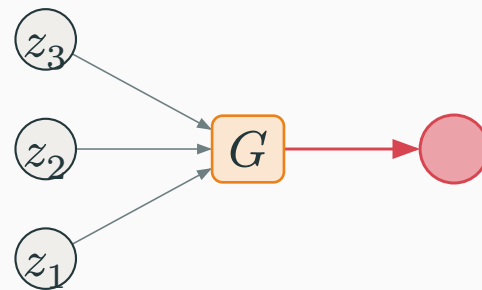
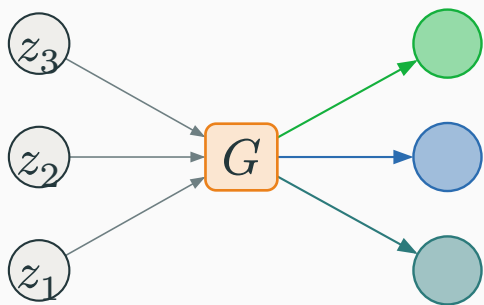
treat DCGAN as an architectural **stabilization milestone**, not a paper to memorize

The generator maps a **simple** connected latent onto a **complicated** data manifold:



interpolation between two codes traces a **path** along this learned map

Send several z values through G and look at the outputs:



left: a **healthy** generator fans out to varied outputs — right: a **collapsed** one maps everything to a single convincing sample

Collapse is the obvious failure; **dropping** is the quiet one:

A generator may cover **several** modes yet **omit rare** regions. The sample grid looks impressive while quietly failing fairness, safety, or scientific-use requirements.

quality and coverage must be evaluated **separately**

GAN losses oscillate and change scale as the opponent changes. Always also inspect:

- **fixed-noise sample grids** over the course of training;
- **diversity** of the generated set;
- **task-specific** downstream measures.

a lower discriminator loss does **not** monotonically mean better samples

Split “quality” into two measurable axes:

Precision — do generated samples lie **near** the data manifold? (fidelity)

Recall — does the generator **cover** the manifold? (diversity)

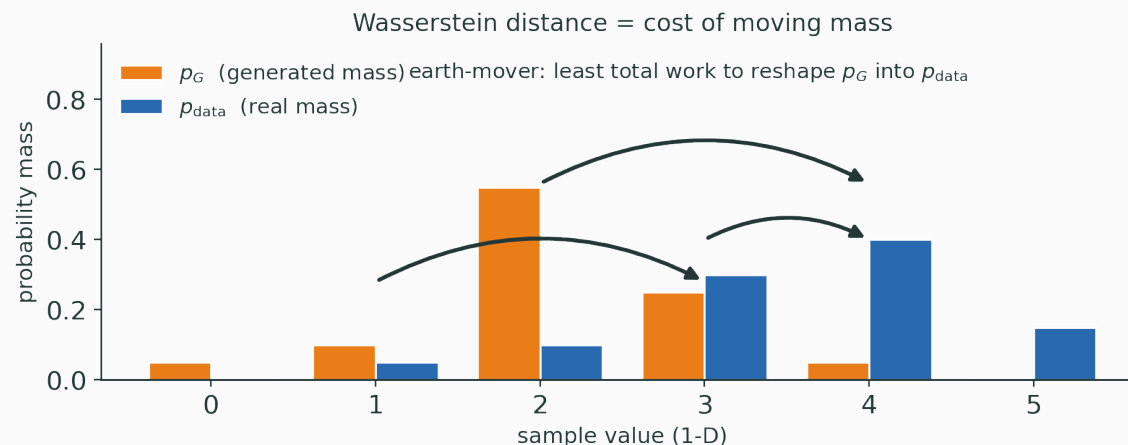
sharp, collapsed outputs can score **high precision** and **poor recall**

Fréchet-style metrics compare **feature statistics** of real and generated images:

one scalar metric $\neq$ complete judgment of quality, diversity, bias, memorization

FID is a **useful summary**, not a verdict. A single number hides mode dropping, bias, and near-copying — inspect samples too.

If two distributions have **disjoint support**, a binary detective becomes overconfident. A distance-like critic measures **how far mass must move**:



Wasserstein critics need **controlled sensitivity** — a bounded slope:

$$|f(x) - f(y)| \leq K \|x - y\|$$

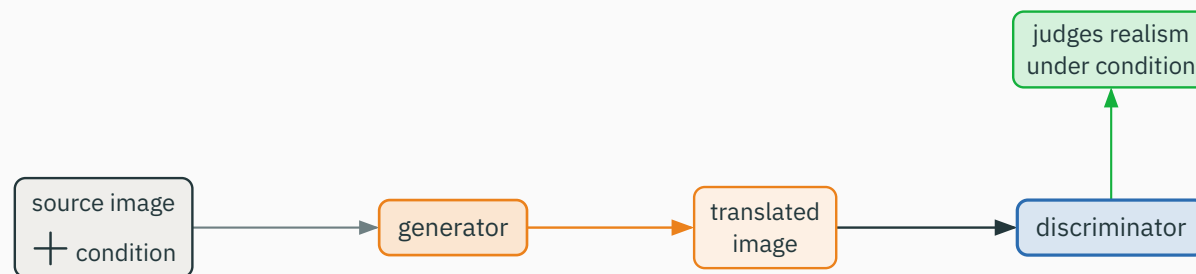
Weight clipping and **gradient penalties** are two ways to enforce this approximate 1-Lipschitz property. We name them, and stop short of deriving WGAN-GP.

Where the latent is **injected** changes what it controls:

- separate latent processing can steer **coarse-to-fine** attributes at **different layers**;
- pose and layout at coarse scales, colour and texture at fine scales.

a clean example of **architecture shaping the meaning** of a latent — not a required implementation

Condition on a **source image** and translate it:



super-resolution · day-to-night · domain adaptation — contrast with diffusion-based editing later

With small or narrow datasets, a generator can **memorize** or nearly copy examples:

A visually persuasive sample grid does **not** by itself establish generalization. Run **nearest-neighbour** checks against the training set.

memorization is a **data-governance** issue, not only a quality one

The discriminator's intermediate features often become **useful representations**:

- separating real from fake requires **semantic** and **texture-sensitive** features;
- those features transfer to downstream tasks — echoing L17.

many generative objectives **incidentally** learn representations — but their evaluation targets differ

Question	VAE	GAN	Diffusion
what trains it?	likelihood bound	adversarial signal	known-noise regression
what is hard?	latent / recon trade-off	game stability	iterative sampling
what we emphasize	probabilistic latent	realism and coverage	denoising dynamics

three routes to the same goal — draw samples that look like the data

Match each **symptom** to a likely issue and one intervention category:

Symptom	Likely issue
fake samples repeat	mode collapse — diversify / rebalance
D accuracy near perfect	D too strong — weaken / critic loss
samples drift around training	oscillation — schedule / regularize
diverse but implausible	under-trained G — more capacity / steps

no symptom has a **unique** fix — name a category, not a magic cure

Misconception	Correction
D is used at generation time	usually only $G(z)$ is needed after training
$D = \frac{1}{2}$ means a bad classifier	it is ideal only when distributions match
sharp samples imply good generation	mode collapse can be sharp and incomplete
GANs maximize likelihood	standard GANs are implicit adversarial models

Test yourself before the next lecture:

1. Why does the generator need gradients **through** a frozen discriminator?
2. What exactly changes in the **non-saturating** loss?
3. How does **mode collapse** differ from poor visual fidelity?
4. Why do **game** losses resist a simple interpretation?

1. A GAN is a **two-player optimization problem**, not a generator plus a conventional label.
2. The discriminator supplies a **learned realism signal**, and its gradients train the generator.
3. Sharp samples do **not** guarantee coverage — **mode collapse** is the central warning.
4. Diffusion solves a **different** training problem: denoising known corruptions, not fooling a moving critic.

A GAN learns a **critic** to tell fake from real; diffusion learns to **reverse a known noise process**.

Next: **Diffusion Models** — a simple corruption process turns generation into **supervised denoising**.